Engineering Mechanics 1

Week 8

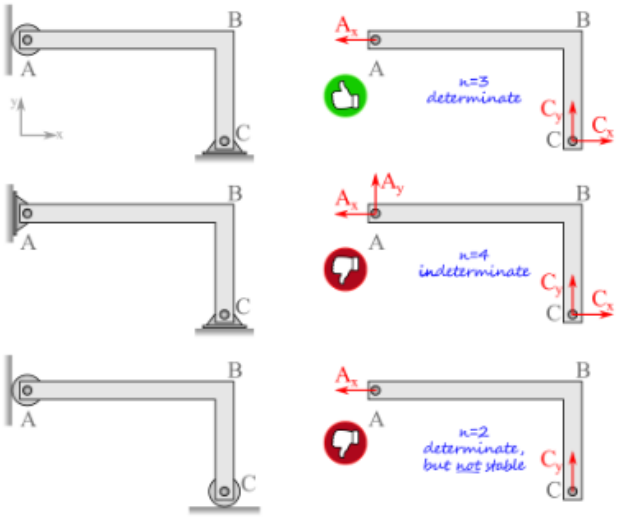
Yuxia Yuan

Autonomous Aerial Systems

Determinacy and Stability

Determinate vs. Indeterminate - CC

A static system is determinate if it is possible to find the unknown reactions using the methods of statics, that is, by using equilibrium equations, otherwise it is indeterminate.



Indeterminate system:

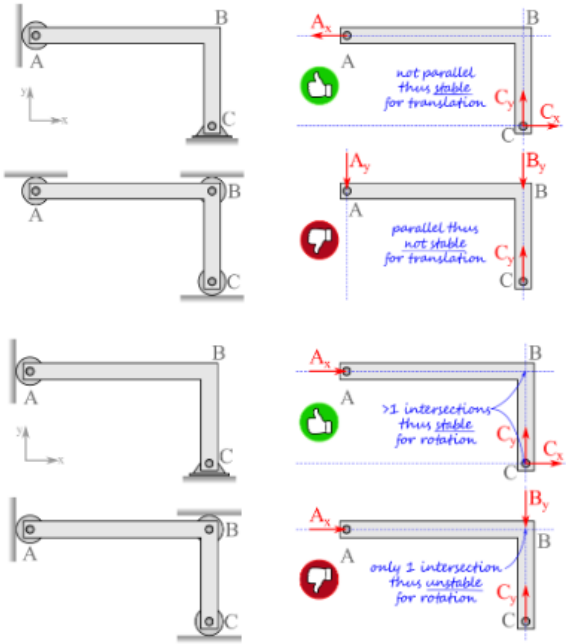
over-constrained: reaction components more than degrees of freedom

under-constrained and **unstable** : reaction components less than degrees of freedom.

Determinacy and Stability

Stable vs. Unstable - PC

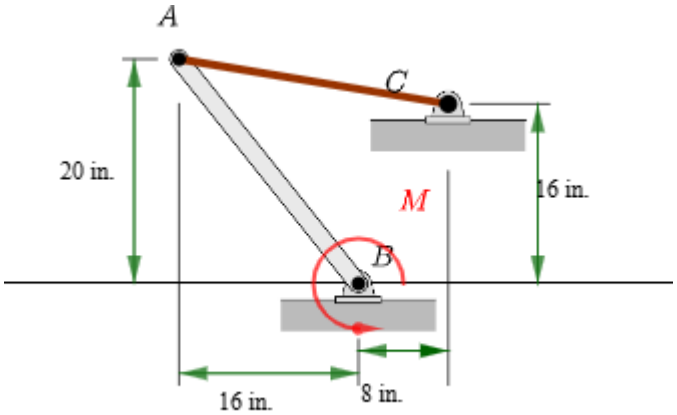
A body in equilibrium is held in **position** by its **supports**, which restrict the body's motion and **counteract the applied loads**



Translation: all the reaction force components parallel to one another

Rotation: lines of action of the reaction forces intersect at a single point

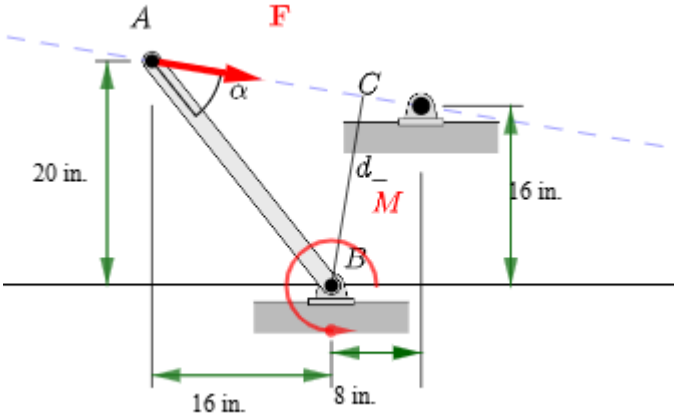
Exercise 1 – 2 mins



Control arm AB is subjected to a counterclockwise moment M and held in position by cable AC

Knowing that the tension in the cable is $15\ lbs$, determine the magnitude of the moment M.

Exercise 1 - Solution



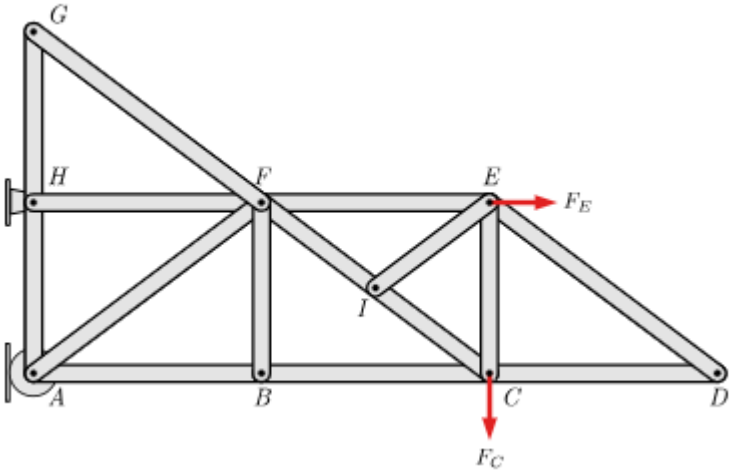
Equate the applied moment M to the moment produced by the force.

$$\Sigma MB = 0 \therefore M = Fd^{\perp}$$

Use geometry to find $d^{\perp}$, then solve for the unknown quantity.

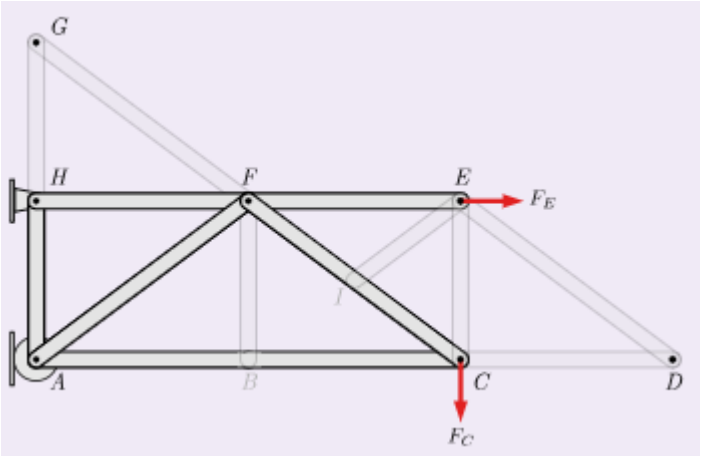
$$M = 256.5\ lbs \cdot in$$

Exercise 2 – 3 mins



Given the truss shown, eliminate all the zero-force members, and draw the remaining truss.

Exercise 2 - Solution

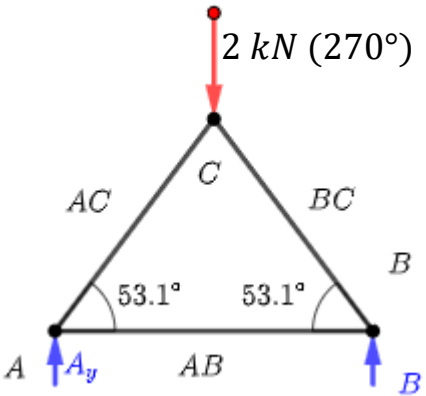


Rule 1: Unloaded node with 2 non-parallel bars: GF, GH, DC, DE

Rule 3: Unloaded node with 3 bars, 2 of which are collinear: BF, IE

Rule 2: Loaded node with 2 bars and force in the direction of the bar: EC

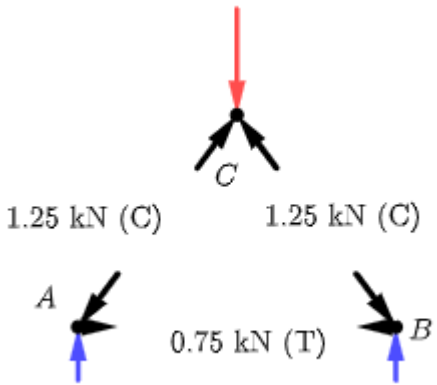
Exercise 3 – 2 mins



The interactive below shows a triangular truss supported by a pin at A and a roller at B, and loaded at joint C.

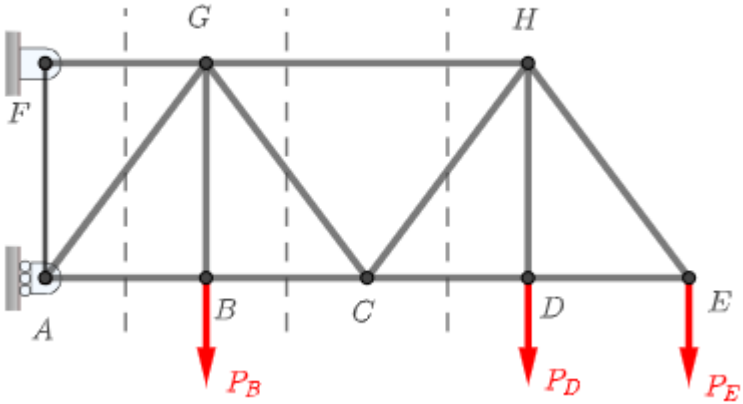
By using Method of Joint to calculate support reactions and internal forces.

Exercise 3 - Solution



Exercise 4 – 5 mins

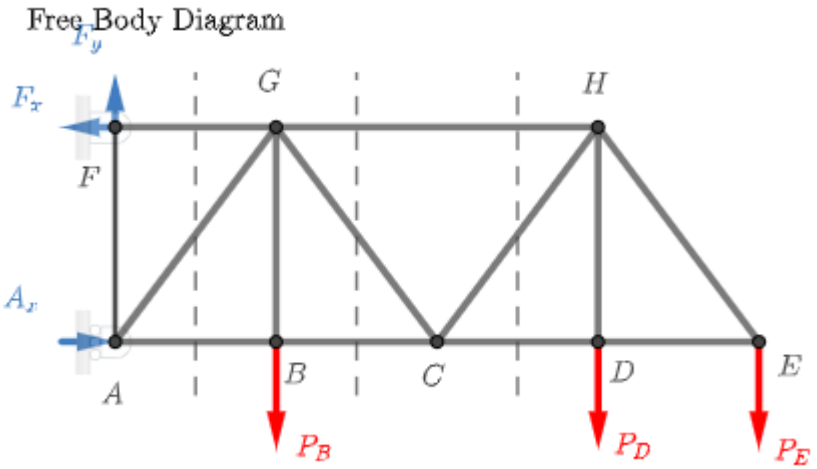
Space Diagram



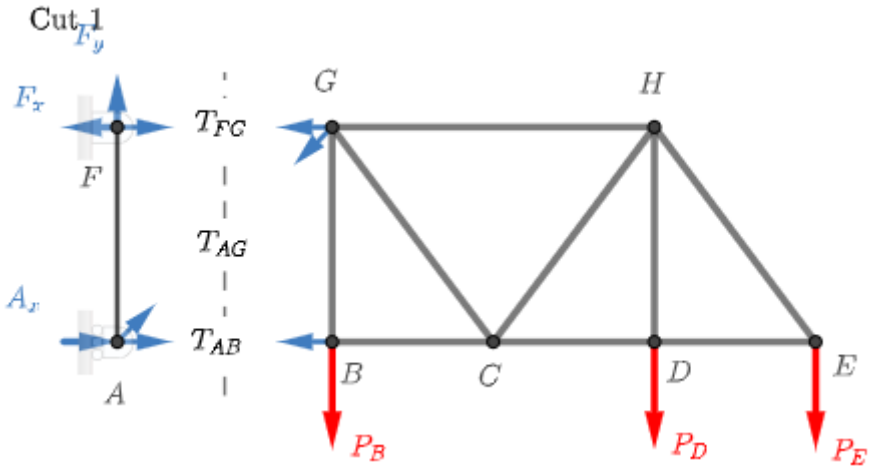
Expose the internal forces in the truss members by cutting through the truss at three locations

The known loads are shown in red, and the unknown reactions at F_x, F_y, A_x, and unknown member forces are shown in blue. Assume all members are in tension.

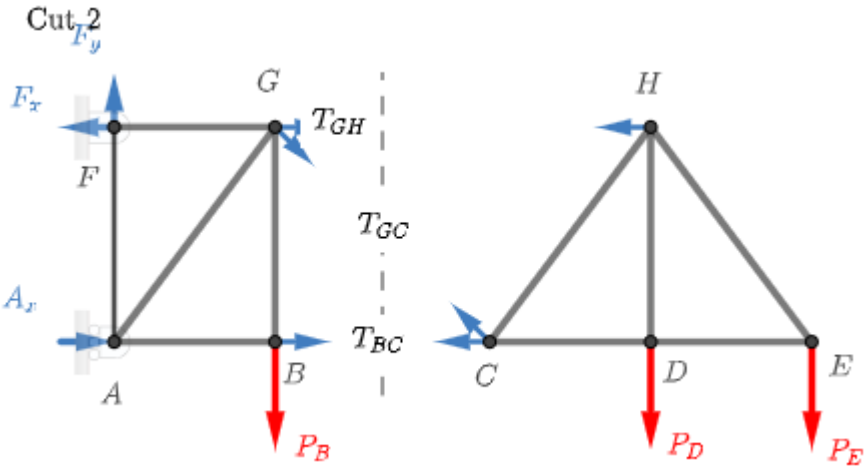
Exercise 4 - Solution



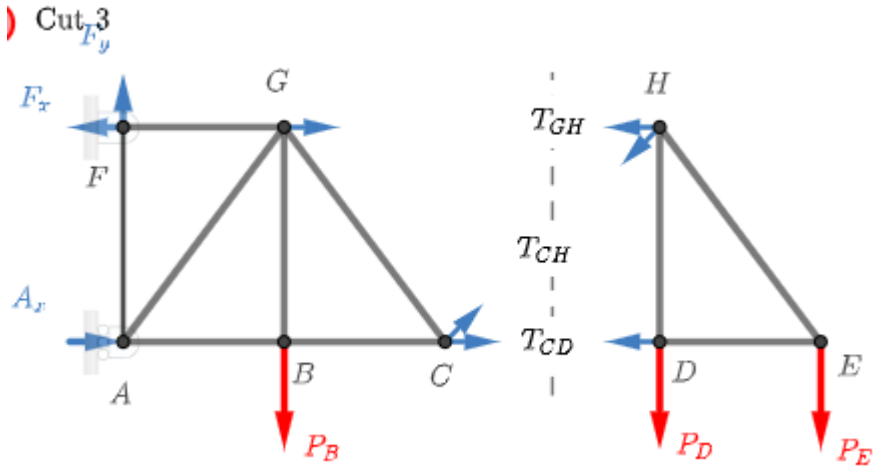
Exercise 4 - Solution



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